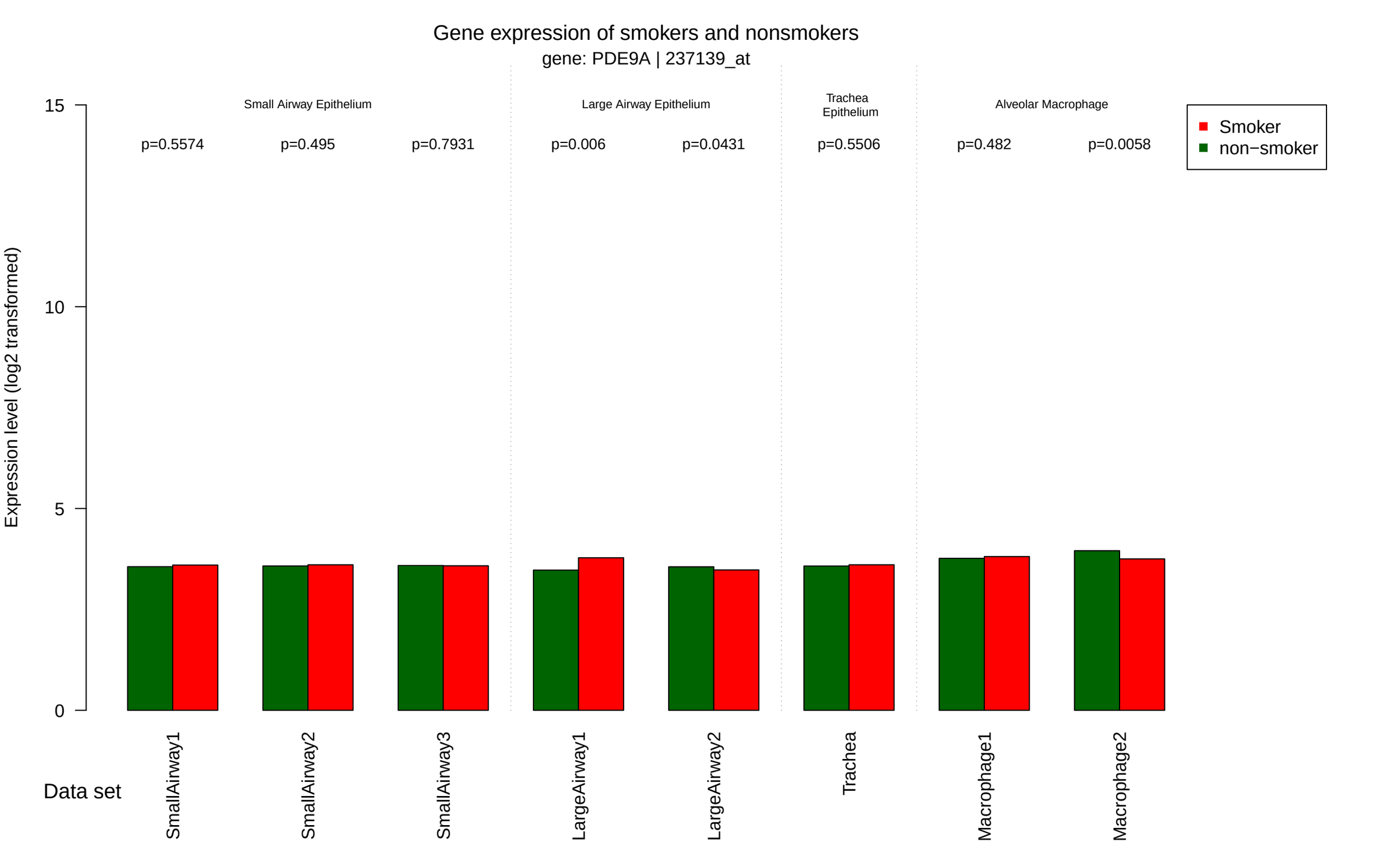
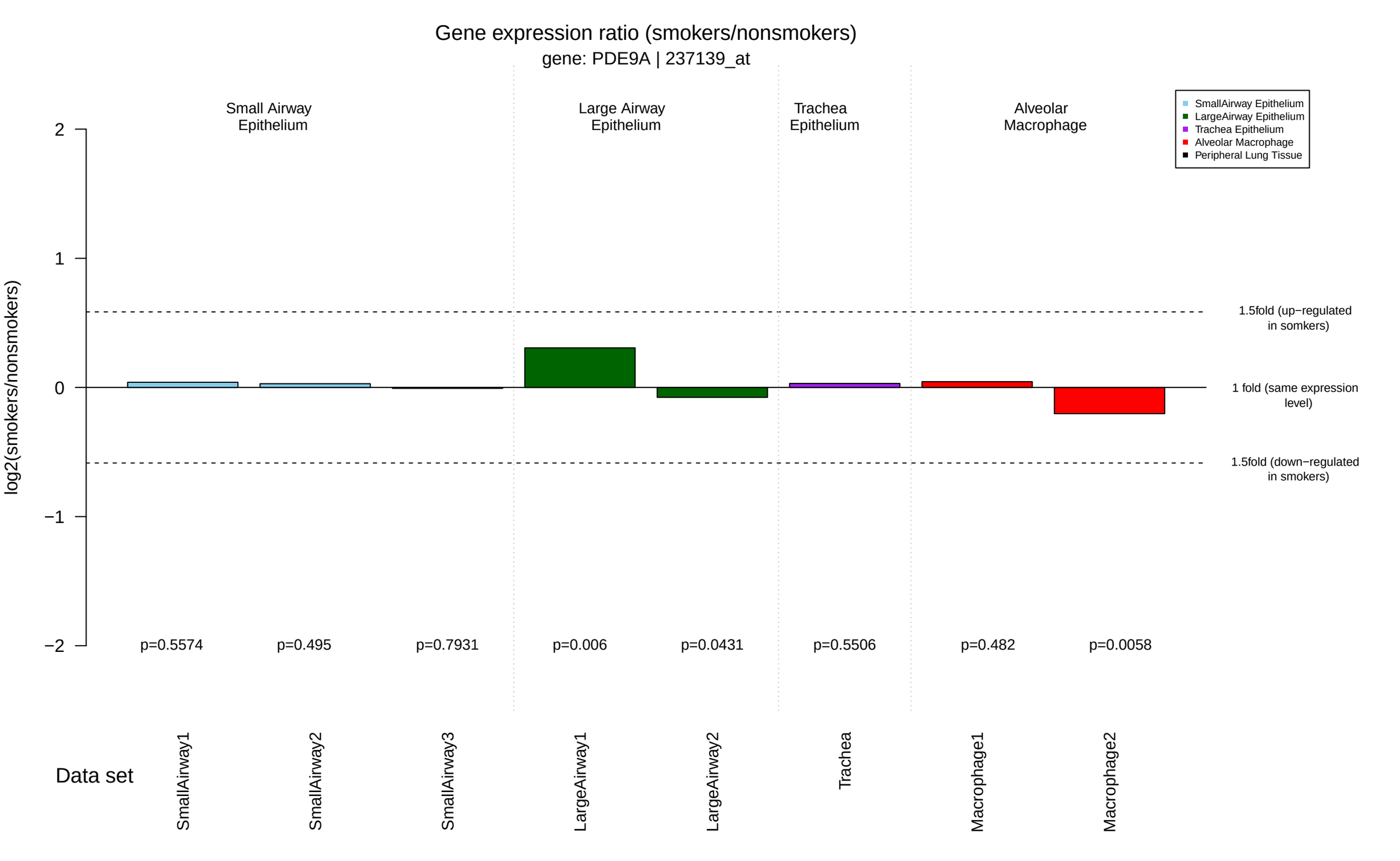
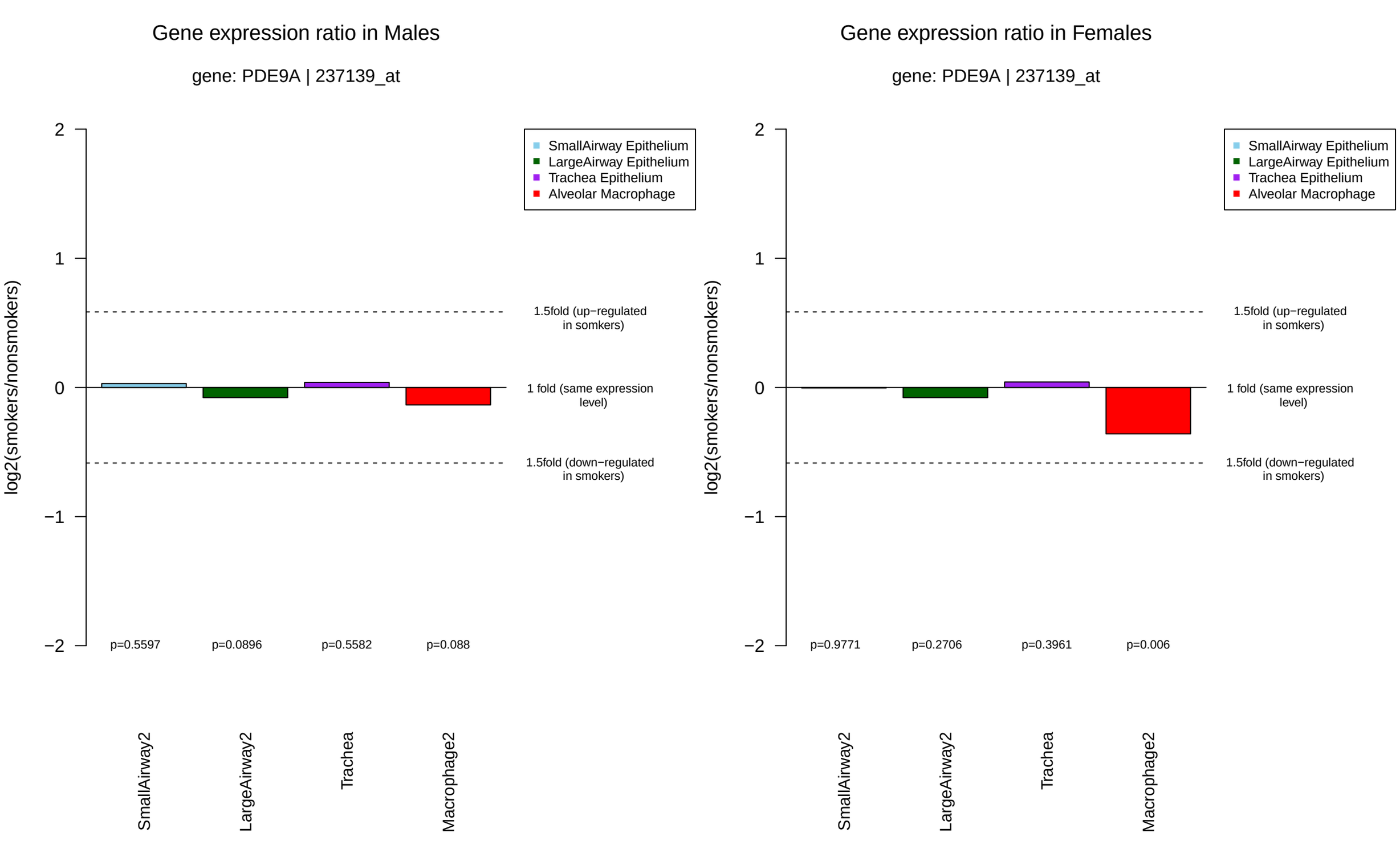
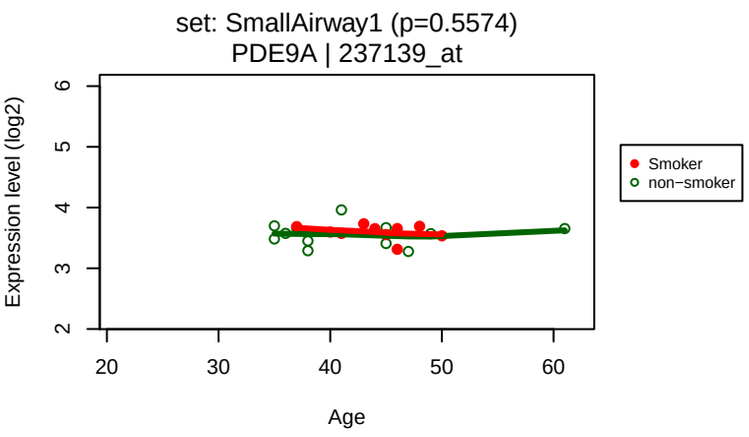
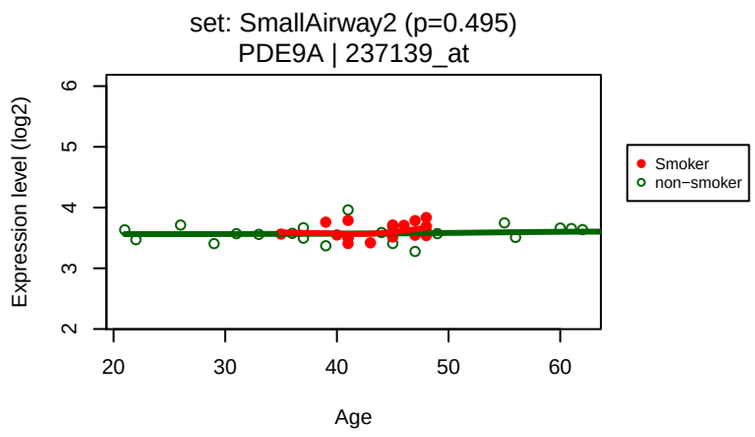
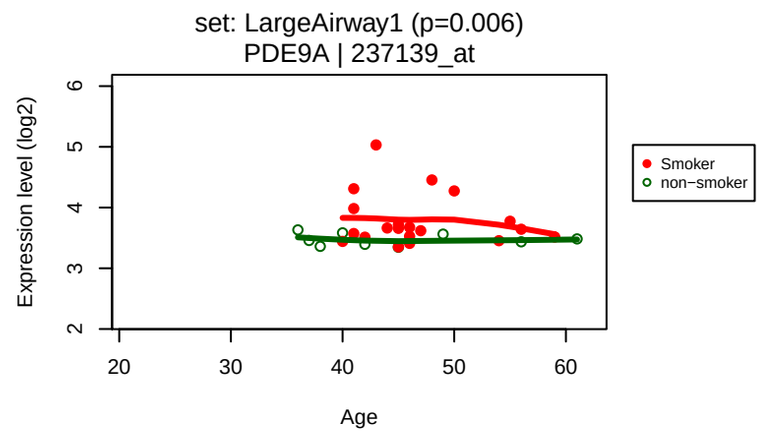

Gene expression of smokers and nonsmokers
gene: PDE9A | 237139_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5574
p=0.495
p=0.7931
p=0.006
p=0.0431
p=0.5506
p=0.482
p=0.0058
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

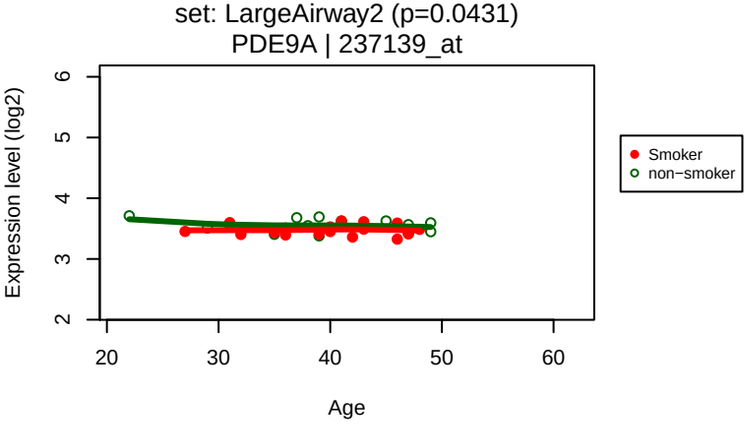

Gene expression ratio (smokers/nonsmokers)
gene: PDE9A | 237139_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5574
p=0.495
p=0.7931
p=0.006
p=0.0431
p=0.5506
p=0.482
p=0.0058
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

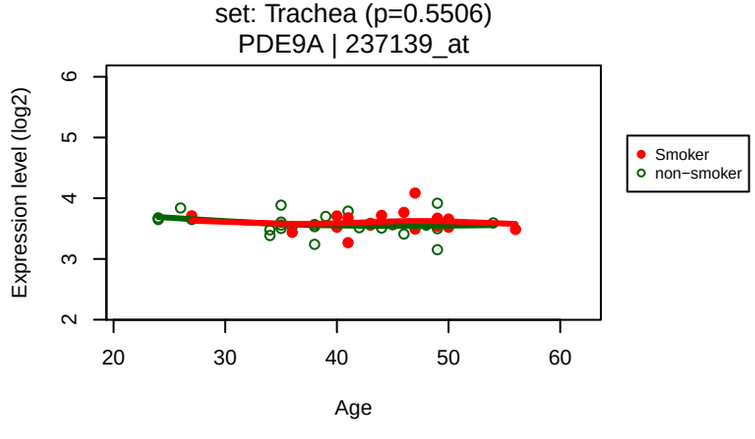

Gene expression ratio in Males
gene: PDE9A | 237139_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5597
p=0.0896
p=0.5582
p=0.088
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PDE9A | 237139_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9771
p=0.2706
p=0.3961
p=0.006
SmallAirway2
LargeAirway2
Trachea
Macrophage2

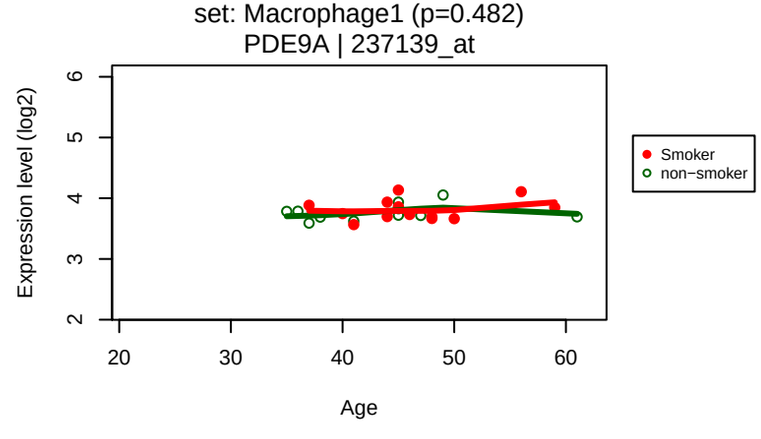

set: SmallAirway1 (p=0.5574)
PDE9A | 237139_at
Expression level (log2)
Age
Smoker
non−smoker

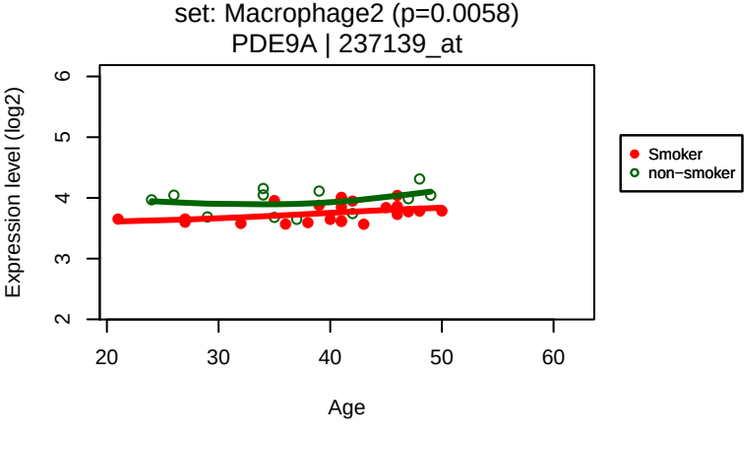

set: SmallAirway2 (p=0.495)
PDE9A | 237139_at
Expression level (log2)
Age
Smoker
non−smoker

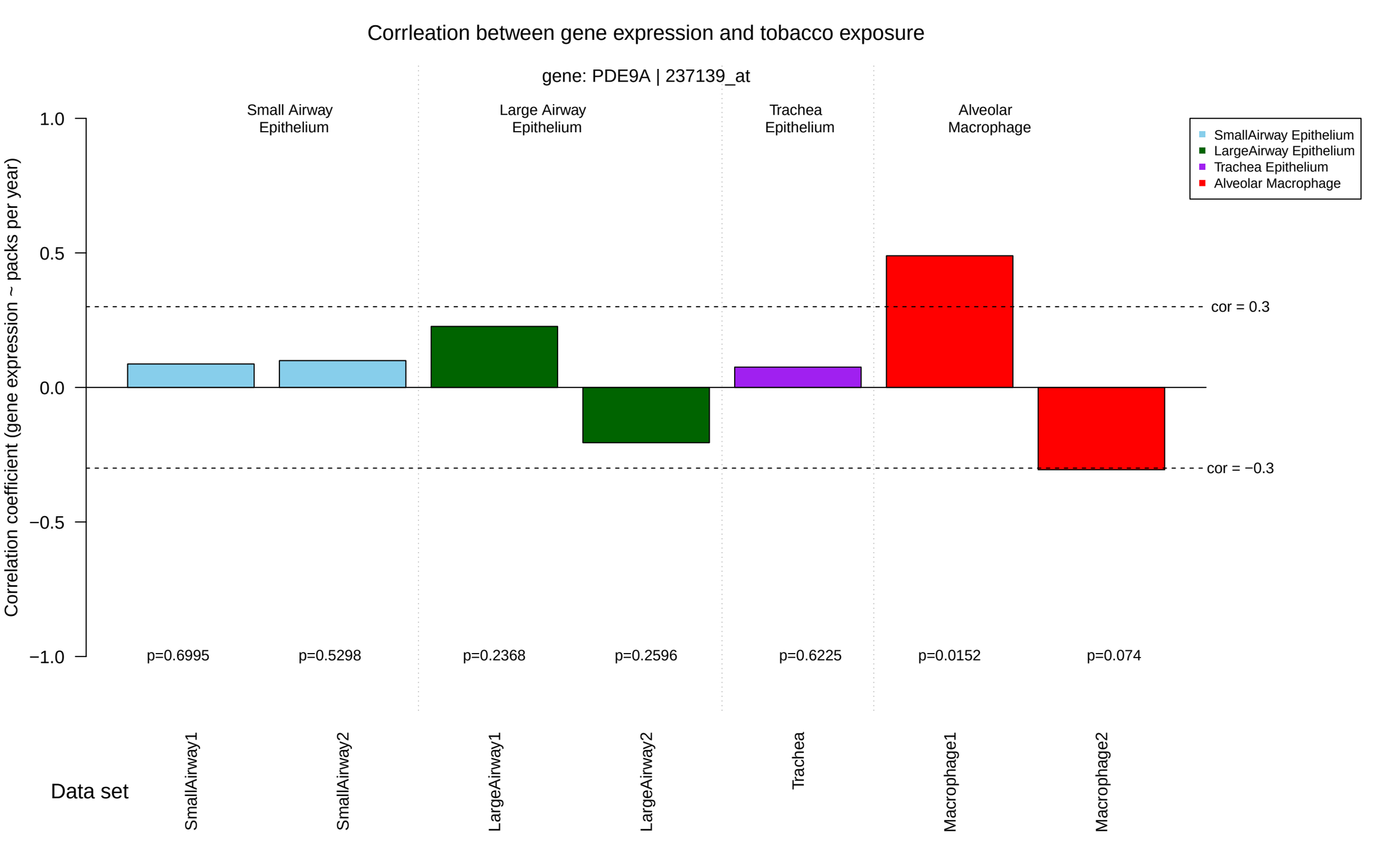

set: LargeAirway1 (p=0.006)
PDE9A | 237139_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0431)
PDE9A | 237139_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5506)
PDE9A | 237139_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.482)
PDE9A | 237139_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0058)
PDE9A | 237139_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PDE9A | 237139_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6995
p=0.5298
p=0.2368
p=0.2596
p=0.6225
p=0.0152
p=0.074
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2